Nuvoton_demo_AEC_16k_0.0.2a Project Notes

1. Project Overview

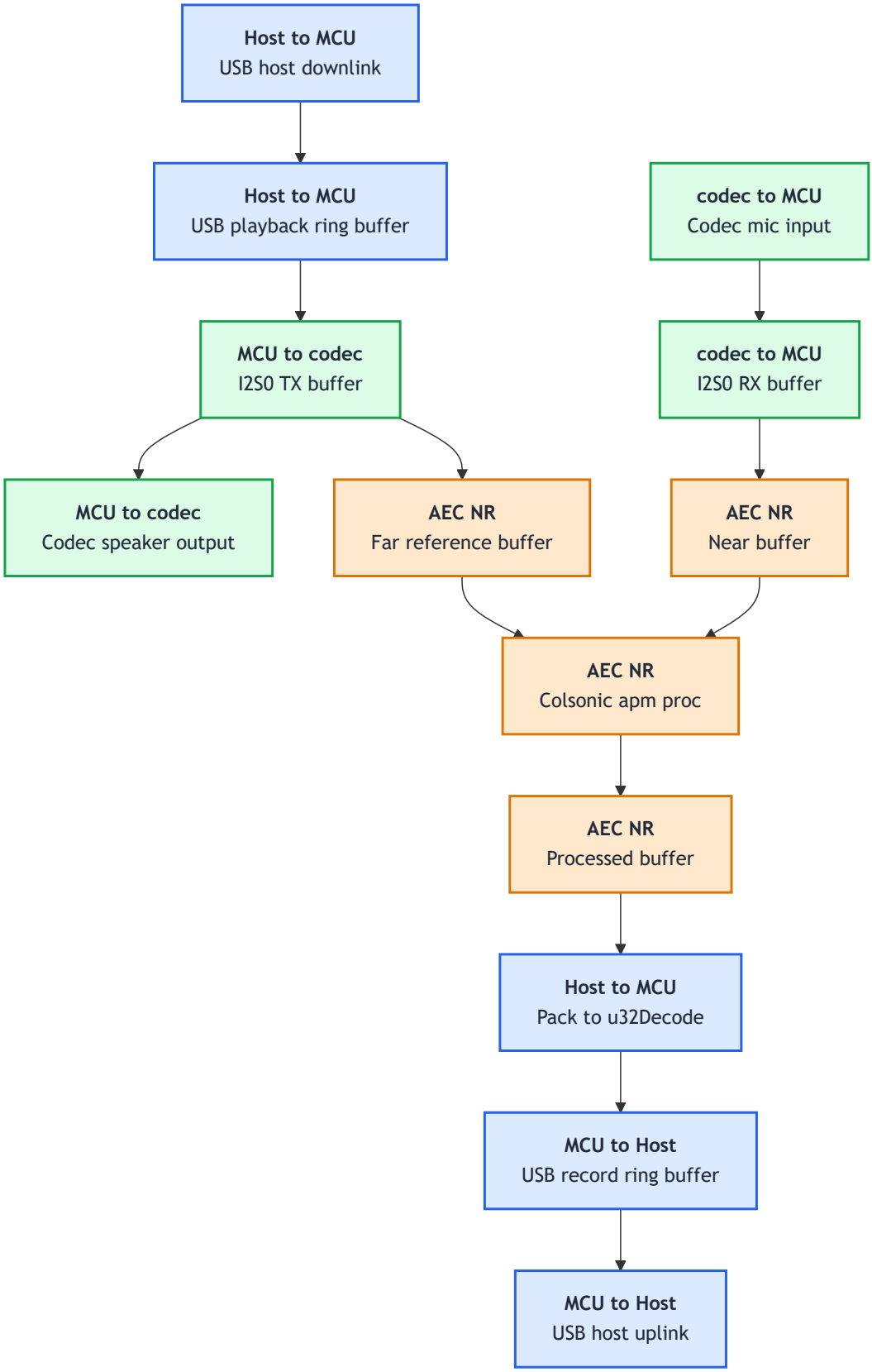
This project is a USB audio + AEC/NR pipeline:

- `USB Host <-> MCU` : UAC playback/record path
- `MCU <-> Codec` : `I2S0` (playback and capture)
- `AEC/NR` : Colsonic APM (`near + far -> processed`)
- Active near-end source in current setting: `I2S0 RX` (`NEAR_END_DMIC = 0`)

2. Audio Data Flow

Read order:

1. USB host downlink audio enters MCU playback ring buffer.
2. MCU sends playback data to codec speaker path, and also builds far reference.
3. Mic capture from codec enters MCU as near signal.
4. AEC/NR uses near + far and generates processed voice.
5. Processed voice is packed and sent to USB host uplink.


Host to MCU
USB host downlink
Host to MCU
USB playback ring buffer
codec to MCU
Codec mic input
MCU to codec
I2S0 TX buffer
codec to MCU
I2S0 RX buffer
MCU to codec
Codec speaker output
AEC NR
Far reference buffer
AEC NR
Near buffer
AEC NR
Colsonic apm proc
AEC NR
Processed buffer
Host to MCU
Pack to u32Decode
MCU to Host
USB record ring buffer
MCU to Host
USB host uplink